FCPL LAB · C# · EVERY SESSION

Every lab, the same way

Keep this page open during the lab. The routine is the same every week; only the brief changes. If you know it already, come in and start.

The brief for each lab, step by step, and what "done" means for it: github.com/x86bogdan/fcpl-2026 → labs. Your theme's plan for every session: your theme's spec → *What each session adds*.

How a lab runs

1 **Come in and start.** You don't have to wait for me. Open today's brief, then get the new tests:

Lab PC (Rider's terminal, `Alt` + `F12`)

```
Download ZIP → extract → Rider
sh sq.sh update
sh sq.sh test
```

Your laptop, with git

```
git pull
sq update
sq test
```

Today's lab shows as *not started* and every earlier lab should still be green. Red on an earlier lab? Fix that first, or ask me for help.

2 **A short briefing.** What today's feature is, one live demo, and what "done" means. Already working? Keep going and listen with one ear: everything on the slides is also in the brief.

3 **Two short huddles at the front.** Join one if it helps, skip both if you know what to do.

What does today mean in my project?
Bring your theme's row (top of this page). You leave with your three lines.

The first step, together
Step 1 of the brief, live on the projector, until your first test runs.

4 **Build.** One step of the brief at a time, `sq test` after each. When you're stuck or green, ask me. Every so often I stop the room for a minute to show a mistake I keep seeing.

5 **Save to GitHub, every time.** Push, or `sh sq.sh upload` and *Upload files*, even if it's red. Work that isn't on GitHub can't be graded, and I can't help with it next week.

Before you write code: three lines

Say what today's feature is *in your app* before you build it. Add a heading for the lab to your repository's `README.md` and write three lines under it: which of your classes change, what they do in your app, and where you'll see it working. Use your theme's row as the starting point. When I verify you, I read these first.

```
## Lab 05 in my project
- ObservationLog becomes the collection: log[0], log["obs-0017"], foreach.
- RecentlyUsed<Telescope> remembers the last 3 telescopes I picked.
- In the app: --recent prints them, newest first.
```

The example is an observatory log, which isn't one of the themes. Yours names your own classes. Can't write the three lines? Join the first huddle: that's what it's for.

Stuck? Climb the ladder

1. **Read the failing test.** `sq test` prints its name and a message that says what's missing. The name is usually the answer to “what does it want?”
2. **Re-read the step** in the brief you're on. Each step says where the code goes.
3. **Look up the syntax** in the C# phrasebook: search with the Java word you know, or paste the compiler error.
4. **Read `reference/`** in your repository for the shape: one theme's solution to the earlier labs.
5. **Ask your neighbour** to explain, not to type it for you.
6. **Ask me for help.** I'll get you unstuck. You still write the code.

AI assistants are allowed if you say so in your README. Ask one to explain an error rather than to write the step: in verification you change and explain your own code.

Ask me: help, or verify

Don't wait for me to come round. Raise your hand or call me over, and say which one you need. I alternate between help and verification, so nobody waits for the whole room.

“Help”	“Verify”	“Late lab”
You climbed the ladder and you're still stuck. I get you moving again in a few minutes. You still write the code.	**Today's lab is green on GitHub.** Green in your repository's **Actions** tab, with every earlier lab still green. Your screen isn't enough: only the CI run counts. If you don't ask, I may not know you're ready.	**An earlier lab you finished after its deadline, or one you missed.** Tell me it's ready. I check late labs when there's time, once help and verification are dealt with.

Getting verified

When I reach you, have open

- your repository's **Actions** tab, on the green run
- your project in Rider (or your IDE), building
- your three lines in the README

About five minutes. I ask for one small change to your code and two “why” questions about it. Tests are 40% of the lab, the change 30%, the questions 30%.

Not reached today? You lose nothing. A lab that's green on GitHub by **Friday 23:59** of its week counts in full, whenever I verify it. After that it's late work, worth up to 8.

Joined late, or missed labs?

Do these in order. Lab 02 comes first because every later lab builds on its types and its store.

1. **Choose a theme** from the catalogue (github.com/x86bogdan/fcpl-2026 → themes) and read its spec. It's your project for the rest of the semester.

2. **Make your repository** from the course template (github.com/x86bogdan/sharpquest-2026): *Use this template*, private, then **Settings → Collaborators → add `x86bogdan`**. The steps are in the template's README.
3. **Claim your avatar** with the claim form. Without it, your results can't reach the scoreboard.
4. **Lab 02.** Follow its brief. `reference/Lab02/` in your repository shows one theme's solution: read it for the shape, then write yours for your theme.
5. **Then join this week's lab in the room.** It counts in full if it's green by Friday 23:59, so don't spend the lab on old ones.
6. **Fill in the other missed labs** at your own pace. Lab 03 can wait: nothing builds on it. If a lab you missed has a catch-up kit in `catchup/`, it lets you carry on, but it earns nothing for that lab.

What a missed lab is worth. Finished later, it's late work: up to 8 points, no stretch bonus, and it doesn't use one of your 4 recoveries. When it's green, tell me it's ready. The full rules are in the course contract (github.com/x86bogdan/fcpl-2026 → CONTRACT.md).

Claim or change your avatar

Your avatar is your name on the scoreboard: one animal in one colour, like **Sky Owl**. The same form is for everything about your entry.

Claim form

- **Missed the form** in session 1: fill it in now.
- **Changed repository** or **want another avatar**: fill it in again with the **same GitHub username**. Your latest answer replaces the earlier one.
- **Repository**: its full address, like `https://github.com/yourname/fcpl-lab`. Not just its name, and not your profile page.
- **Invited `x86bogdan` more than a week ago** and nothing happened? Invitations expire after 7 days. Send it again: Settings → Collaborators.

Open the claim form (forms.gle/E5Us5KN8srhMMMEH8)

Which avatars are already taken: the live list on this page online, x86bogdan.github.io/fcpl-2026/routine#avatar.

Everything else

Lab briefs
Each lab step by step, and what “done” means
(github.com/x86bogdan/fcpl-2026/tree/main/labs)

C# phrasebook
C# next to Java and C++, and the compiler errors
(x86bogdan.github.io/fcpl-2026/cheatsheet)

Course contract
Grading, late work, recovery, AI
(github.com/x86bogdan/fcpl-2026 → CONTRACT.md)

Working on a lab PC
No git: ZIP, Rider's terminal, upload
(github.com/x86bogdan/fcpl-2026 → LAB-PC.md)

The 21 themes
Each with its plan for every session
(github.com/x86bogdan/fcpl-2026 → themes)

Leaderboard
Updated after each week's labs
(x86bogdan.github.io/sharpquest-2026)